

PRINCIPLES OF FLIGHT

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BASIC AERODYNAMIC THEORY



LIFT EQUATION

QUIZ !!

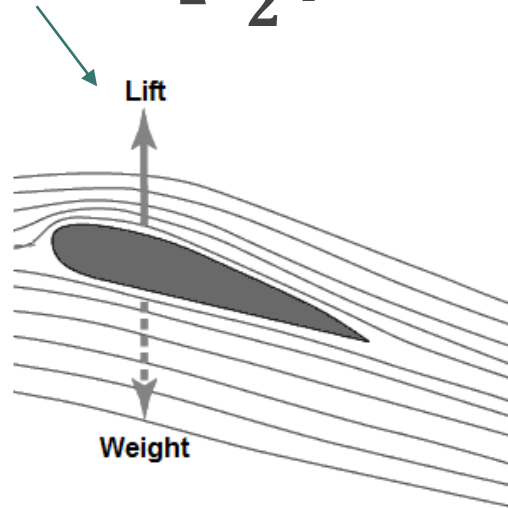
1. What is the difference between force, pressure, and moment? Include their units.
2. What happens to air density, air pressure, and air temperature as we go higher in altitude?
3. What control surface on an airplane causes yaw movement? Around which axis does this movement happen?
4. What happens to the speed of air flow and static pressure when the area it moves through gets smaller? What theories is this based on?
5. What is the angle of attack?

LIFT EQUATION

$$L = C_L \left(\frac{1}{2} \rho v^2 \right) S$$

LIFT FORCE

$$L = C_L \left(\frac{1}{2} \rho v^2 \right) S$$

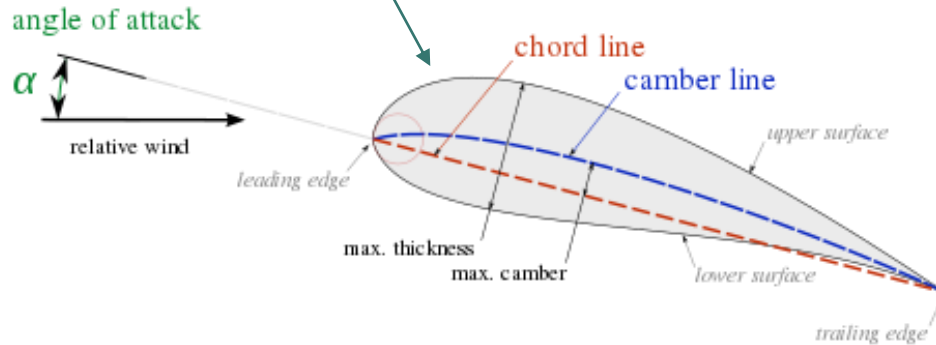


LIFT COEFFICIENT

Depends on :

- Airfoil Geometry (camber, LE, TE, etc)
- Angle of Attack

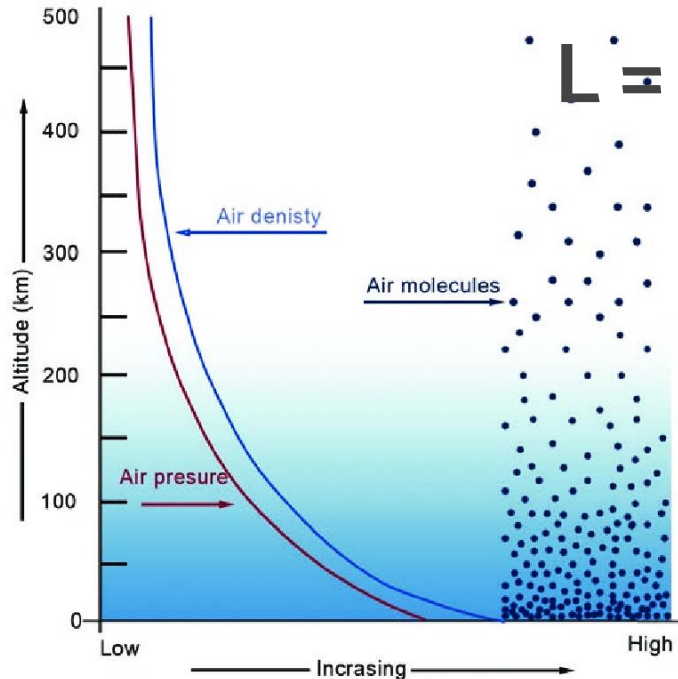
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DYNAMIC PRESSURE

Depends on :

- Air Density (flight altitude)
- Airspeed



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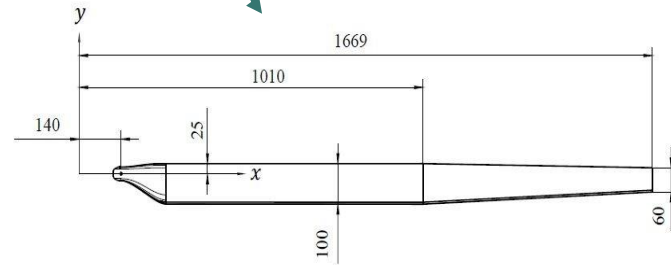
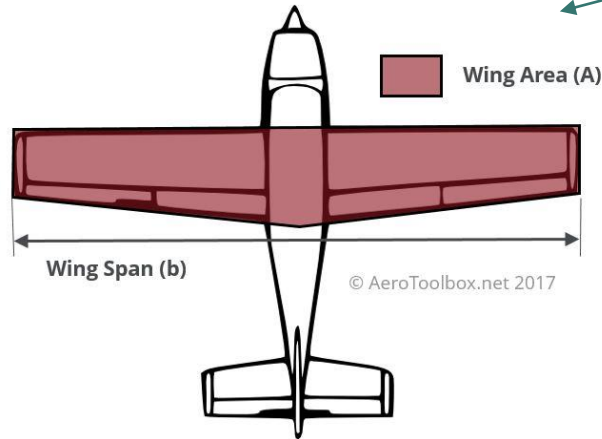


WING/BLADE AREA

Depends on :

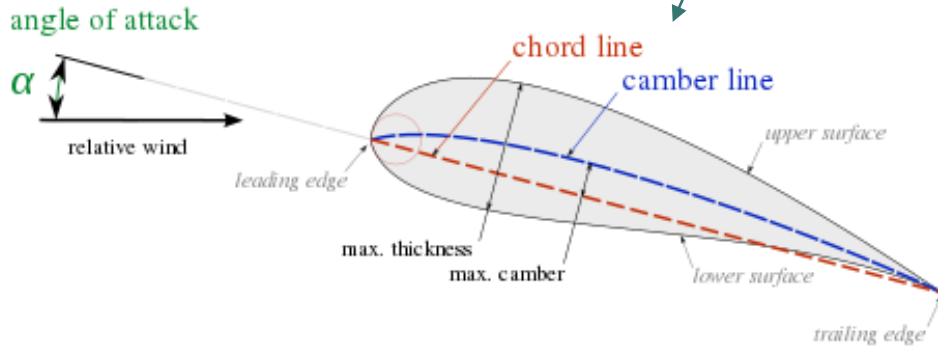
- Wing area for fixed wings
- Rotor blade Area for helicopters

$$L = C_L \left(\frac{1}{2} \rho v^2 \right) S$$



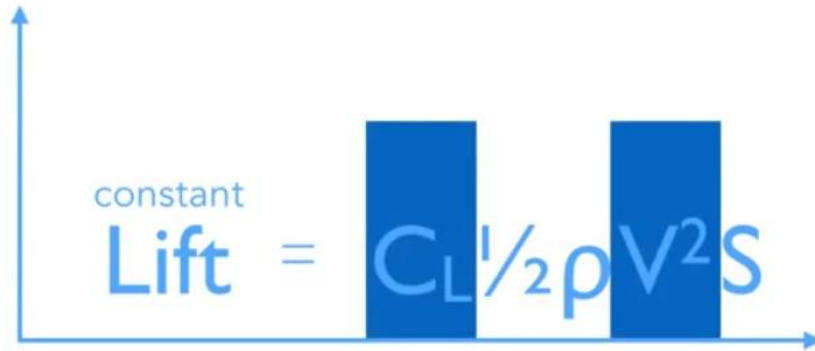
LIFT FORMULA

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LIFT FORMULA

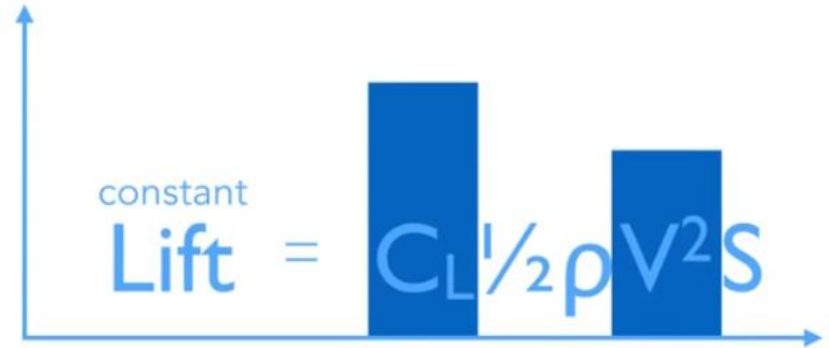
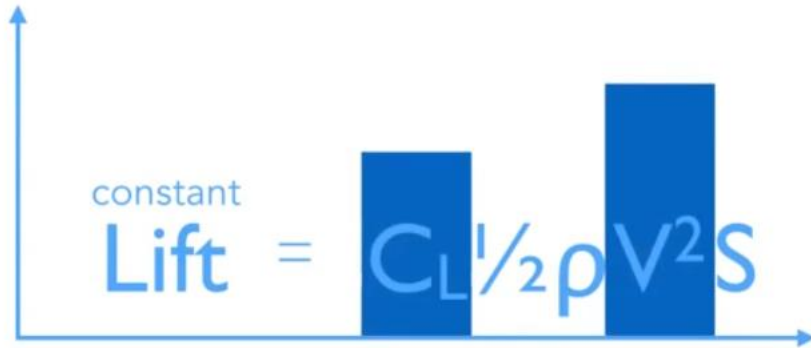
- The lift formula can also be used to consider the relationship between speed and angle of attack at a constant altitude (air density)
- As speed is changed, angle of attack must be adjusted to keep lift constant.



constant
Lift = $C_L \frac{1}{2} \rho V^2 S$

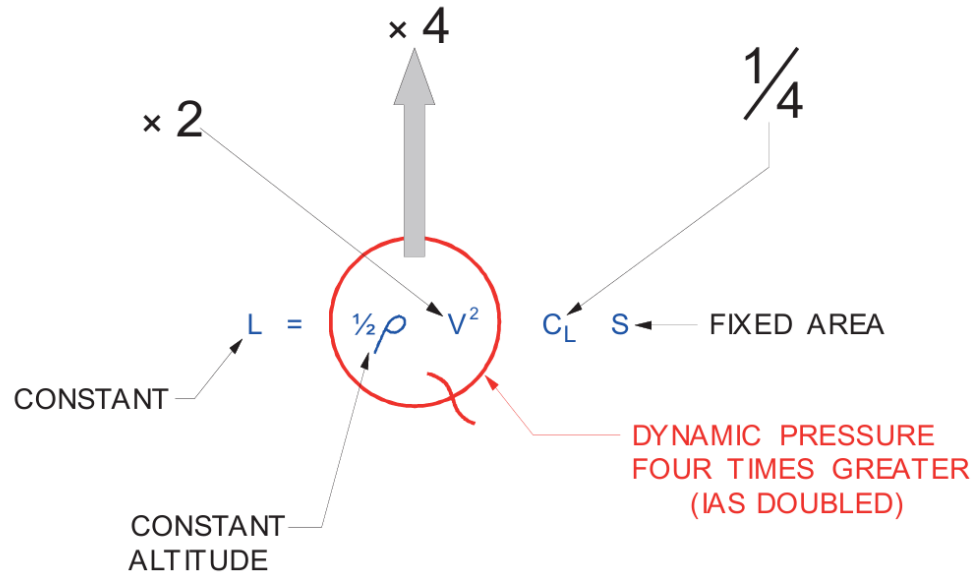
LIFT FORMULA

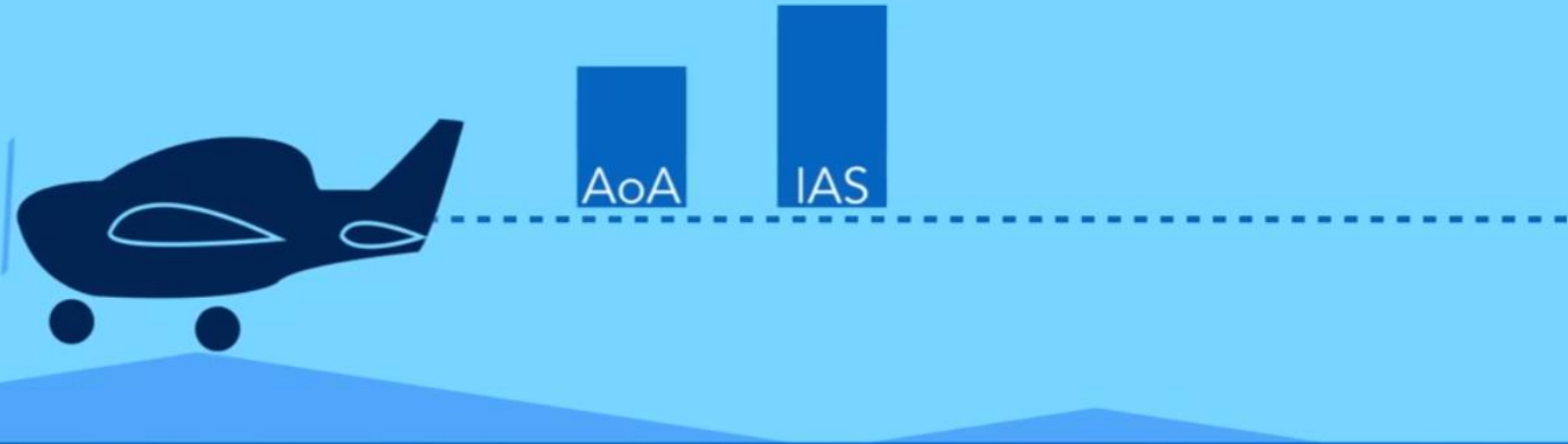
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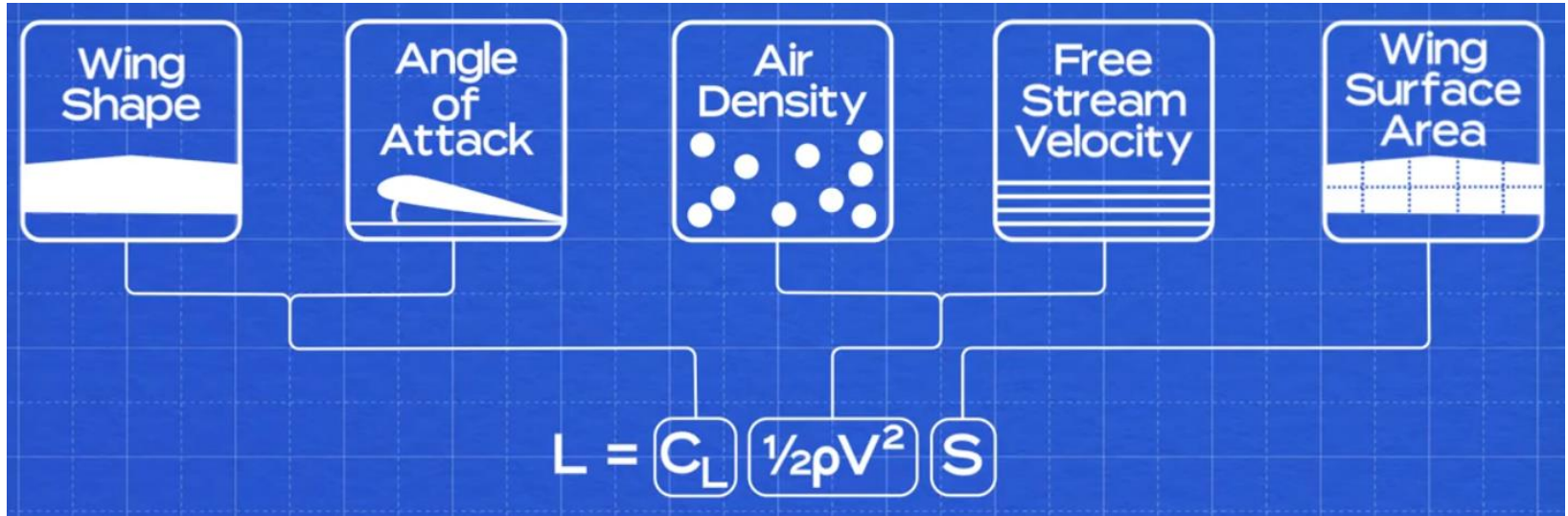
LIFT FORMULA

IF SPEED IS DOUBLED, C_L MUST BE REDUCED
TO $\frac{1}{4}$ OF ITS PREVIOUS VALUE

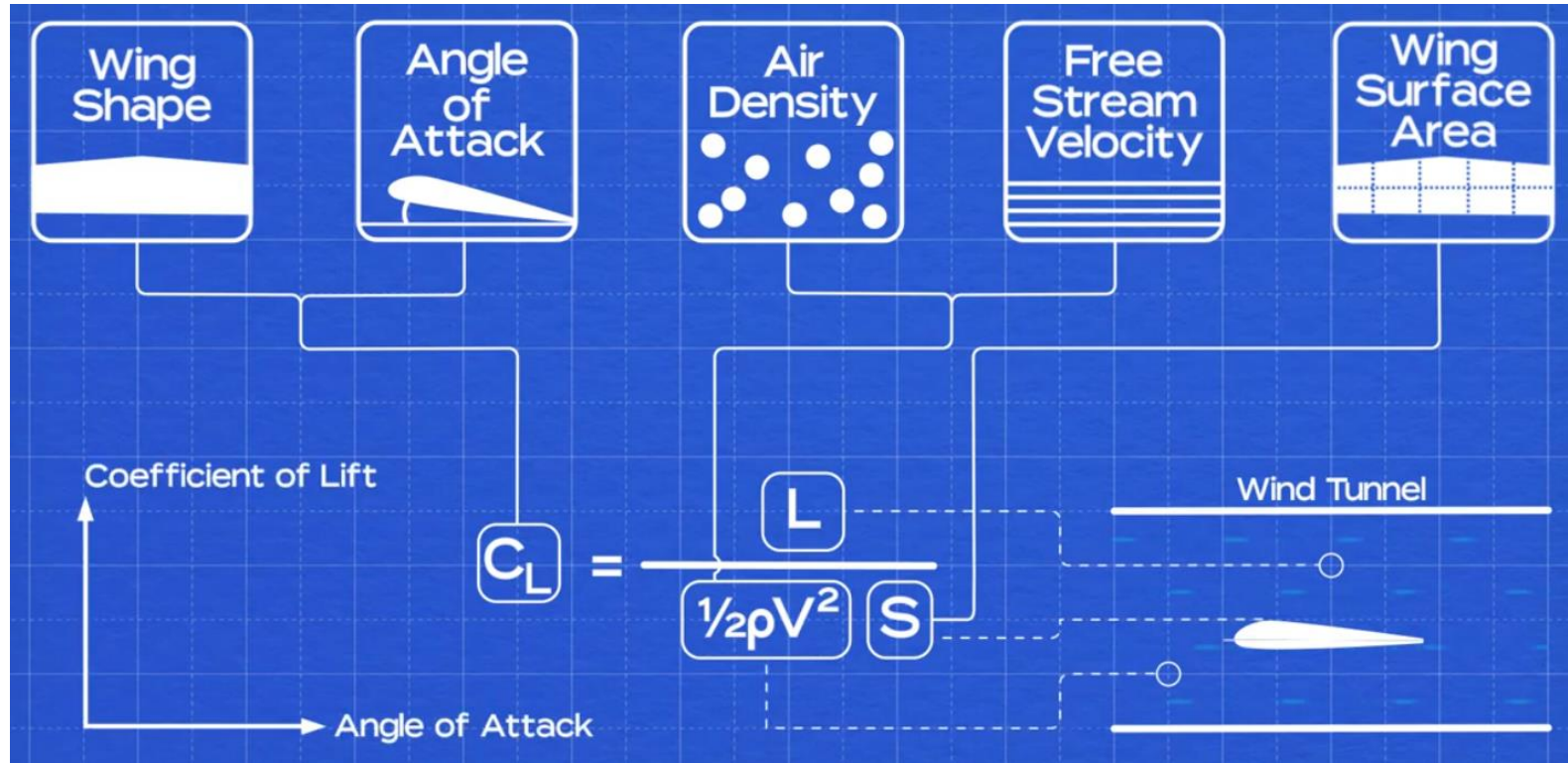




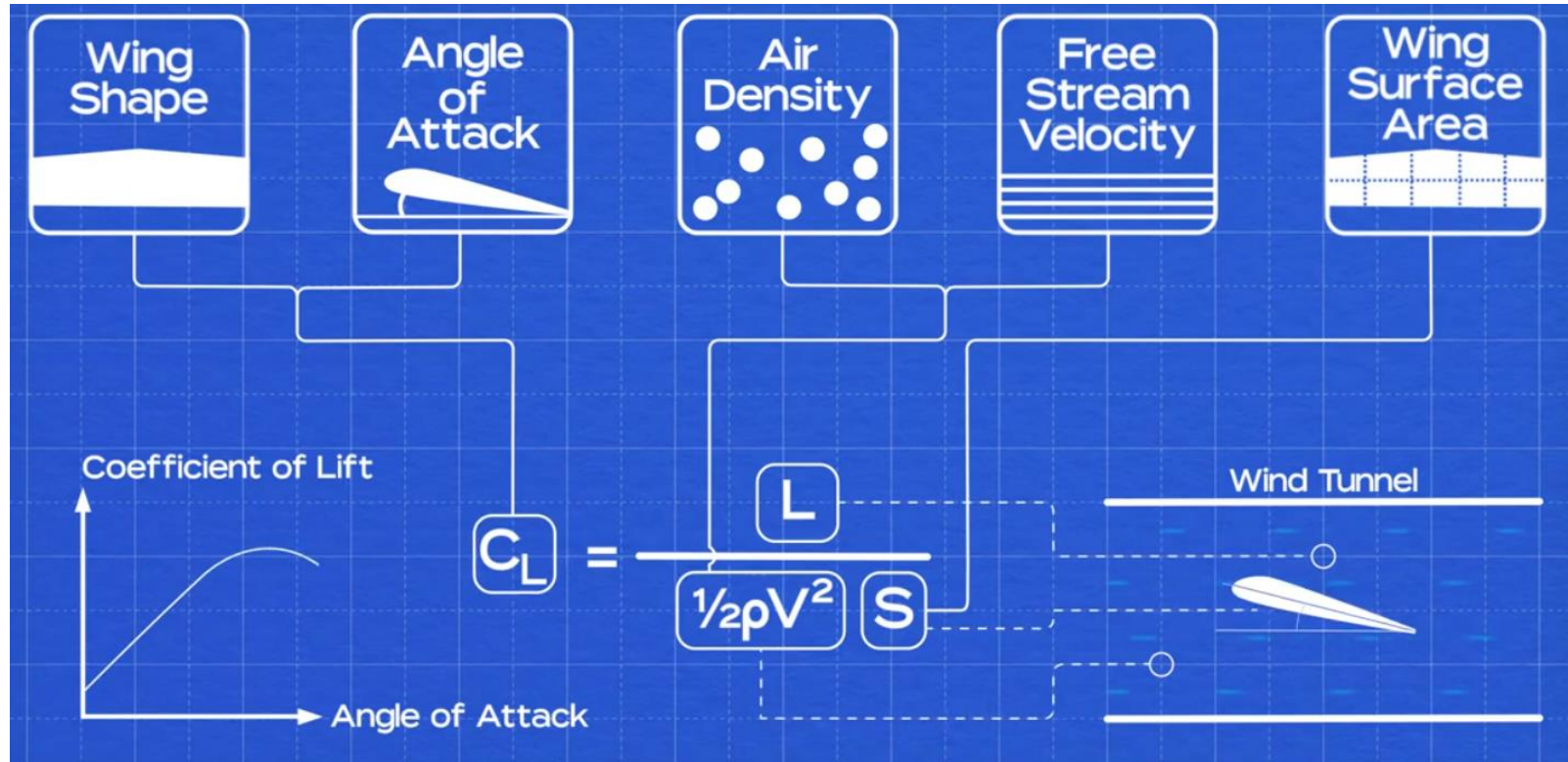
LIFT FORMULA



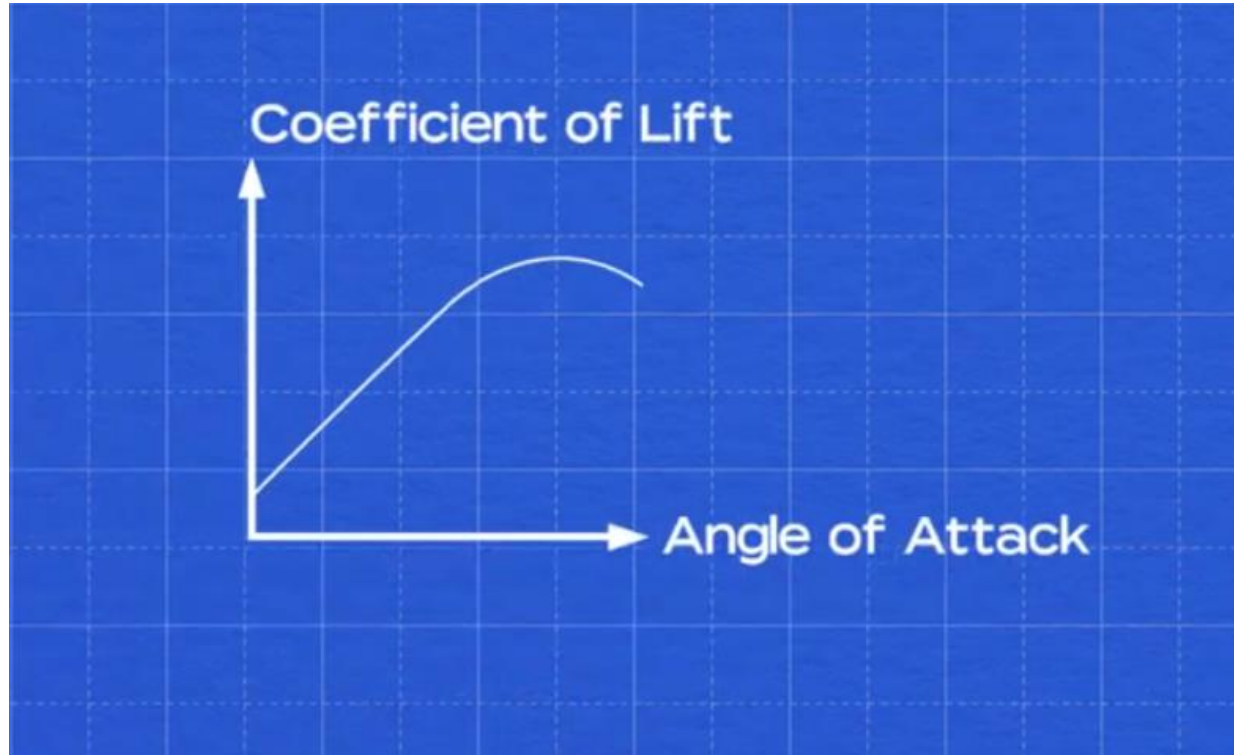
LIFT COEFFICIENT



LIFT COEFFICIENT

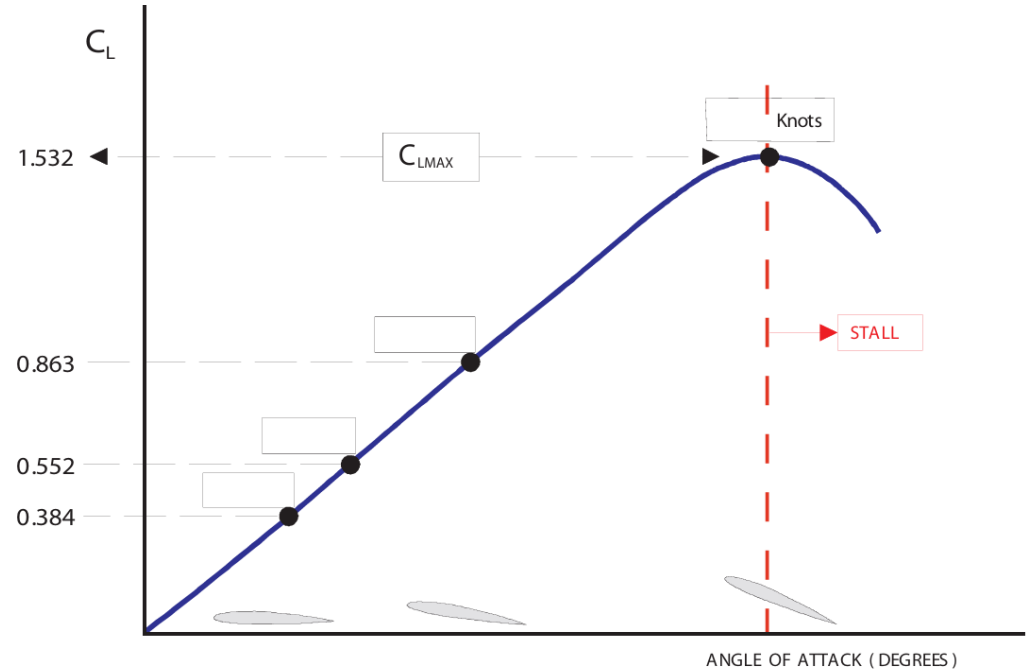


LIFT COEFFICIENT



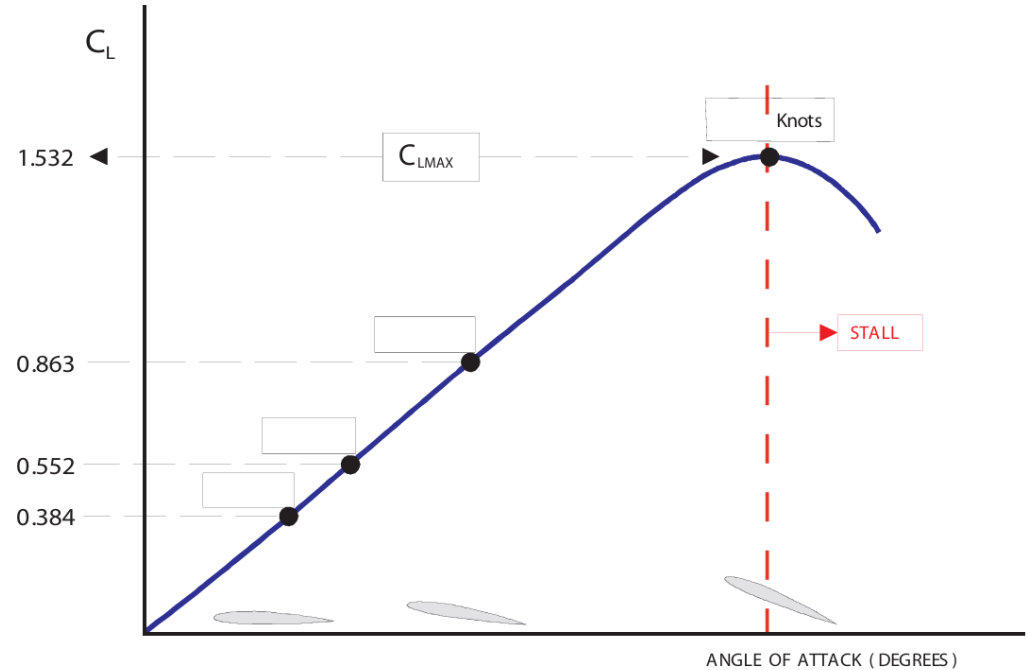
LIFT COEFFICIENT

- lift coefficient (C_L) plotted against angle of attack. It is evident that the section is symmetrical because no lift is produced at zero angle of attack.
- Lift coefficient increases with angle of attack up to a maximum (C_{LMAX}), which corresponds to the “Critical” angle of attack. Continuing to increase the angle of attack beyond this point makes it impossible for the airflow to maintain its previous smooth flow over the contour of the upper surface, and lift will reduce. This phenomena, stall, will be discussed in detail later.



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